INCH-POUND

MIL-T-82839(OS)
5 February 1990
SUPERSEDING
WS 20884
25 October 1983
(See 6.6)

MILITARY SPECIFICATION

TAPE, CARBON CLOTH, PHENOLIC-PREIMPREGNATED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification defines the requirements for one type of carbon cloth preimpregnated with phenolic resin, referred to herein as the tape.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards, and handbooks.** The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-R-9299	Resin, Phenolic, Low-Pressure Laminating
MIL-B-82792	Broadgoods, Carbon Cloth Phenolic-Preimpregnated

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self addressed Standardization Document Improvement Proposal (DD Form 1423) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:28Z
Check the source to verify that this is the currentversion before use.

MIL-R-82817 Reinforcement, Carbon

MIL-T-82846 Thread, Splicing, Nylon

STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C 177 Thermal Conductivity of Materials by Means of the Guarded Hot Plate Apparatus (DoD adopted)

ASTM D 695 Compressive Properties of Rigid Plastics

ASTM D 740 Methyl Ethyl Ketone (DoD adopted)

ASTM D 790 Flexural Properties of Plastics

ASTM E 228 Linear Thermal Expansion of Rigid Solids with a Vitreous Silica Dilatometer, Test for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 **Material.** The material shall be in the form of tape rolls cut from woven carbon cloth impregnated with carbon-filled phenolic resin in accordance with MIL-B-82792. The tape shall be cut on the bias, or 45 $\pm$ 5 degrees from the warp axis, and butt spliced by sewing with nylon thread which conforms to MIL-T-82846 (see 6.2).

3.2.1 **Resin.** The resin system shall conform to the requirements of MIL-R-9299, Grade A (see 6.2).

3.2.2 **Reinforcement.** The reinforcement shall be a carbon fabric conforming to MIL-C-82817 (see 6.2).

3.2.3 **Resin filler.** The resin filler shall be carbon black with a minimum carbon assay of 90 percent (see 6.2).

3.3 **Properties.**

3.3.1 **Uncured material.** The properties of the uncured tape shall be in accordance with table I.

3.3.2 **Cured material.** The properties of the cured tape shall be in accordance with table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

TABLE I. Uncured material physical properties.

Property	Percent by Weight	
	Minimum	Maximum
Cloth content	47.0	63.0
Resin solids, dry weight	32.0	37.0
Filler content	5.0	16.0
Volatile content	4.0	6.5
Resin flow	10.0	20.0

TABLE II. Physical and mechanical properties of cured material.

Property	Limits	
	Minimum	Maximum
Density, g/cc	1.44	1.55
Residual volatiles, percent	0.0	2.50
Resin content, percent	30.0	38.5
Compressive strength, psi		
Warp direction	30,000	65,000
Fill direction (edgewise)	20,000	55,000
Interlaminar double shear strength, psi	2,400	
Thermal conductivity, Btu/(hr)(ft) degrees Fahrenheit (°F) (250°F)		
With ply - fill direction		1.1
Across ply		0.9
Coefficient of thermal expansion, in./in.-°F x 10^{-6} at 75 to 400°F		
Across ply	5.0	20.0
With ply	2.0	8.0
Flexural strength, psi		
Warp direction	25,000	55,000
Fill direction	20,000	50,000

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

3.4 **Tape dimensions.** Tape width tolerance shall be ± 1/8 inch. The tape shall be wound on cores with an inside diameter of 3.0 inches. The outside diameter of the tape rolls shall not exceed 20 inches.

3.5 **Age of material.** The tape shall be shipped within 3 months after manufacturing date of the broadgoods and shall be tested and certified within 60 days of shipment. The contractor shall certify that the material has been stored at a maximum temperature of 50°F between date of manufacture of the broadgoods and date of shipment (see 6.2). Temperature excursions above 50°F are permissible provided the cumulative excursions do not exceed 30 days.

3.6 **Shelf life.** The tape specified herein shall have a shelf life of 9 months after date of manufacture of broadgoods material when stored at 50°F maximum in sealed watertight plastic bags. The cumulative out of storage time shall not exceed 30 days.

3.6.1 **Shelf life extension.** The shelf life of all tape rolls within a single storage container may be extended for one additional 6 month period provided the average value of three specimens from the beginning of one tape roll with the longest out of storage time from the storage container conforms to the following when tested as specified in 4.6.2.1 and 4.6.2.3.

	Minimum	Maximum
Resin flow, percent by weight	5.0	25.0
Volatile content, percent by weight	2.5	6.5

3.7 **Workmanship.** The tape shall have no holes, tears, non-uniform weave, and shall be free from dirt, grease, oil, and other contaminants which may render it unsuitable for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the tests and inspections specified in tables III and IV. First article data obtained for broadgoods supplied to MIL-B-82792 may be used to show conformance to the first article requirements of this specification. Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

TABLE III. First article inspection.

Tests and Inspections	Requirement	Test Method
Cured Material		
Residual volatiles	3.3.2	4.6.3.1.2
Compressive strength	3.3.2	4.6.3.1.4
Thermal conductivity	3.3.2	4.6.3.1.6
Coefficient of thermal expansion	3.3.2	4.6.3.1.7
Flexural strength	3.3.2	4.6.3.1.8

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table IV.

4.4.1 **Sampling.**

4.4.1.1 **Lot.** A lot shall consist of all broadgoods rolls which are impregnated in one continuous production run, using one batch of the impregnator's resin and additives mixed in one vessel at one time, and subject to inspection at one time.

4.4.1.2 **Sampling for uncured material properties.** Each broadgoods roll in the lot shall be tested for uncured material properties at the beginning and at the end of the roll. Three specimens shall be tested for each uncured material property.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

TABLE IV. Quality conformance inspection.

Tests and Inspections	Requirement	Test Method
Uncured Material		
Visual examination	3.2, 3.5, 3.7 and section 5	4.6.1
Volatile content	3.3.1	4.6.2.1
Filler content	3.3.1	4.6.2.2
Resin solids, dry weight	3.3.1	4.6.2.2
Cloth content	3.3.1	4.6.2.2
Resin flow	3.3.1	4.6.2.3
Cured material		
Density	3.3.2	4.6.3.1.1
Resin content	3.3.2	4.6.3.1.3
Compressive strength (warp)	3.3.2	4.6.3.1.4
Interlaminar shear	3.3.2	4.6.3.1.5

4.4.1.3 **Sampling for cured material properties.** Sampling for cured material properties tests shall be in accordance with table V. Three tests shall be performed at the beginning of each sampled roll end.

TABLE V. Broadgoods roll sampling for cured material properties.

Number of Broadgoods Rolls in Lot	Number of Sample Rolls to be Selected for Testing
1 to 3	ALL
4 to 40	3

4.5 **Test results.** The average results of three tests at each location for each property shall conform to the specified property requirements in section 3. Individual specimen test results shall also be recorded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

4.6 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the test or procedure description, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined in this specification. Solvents and indicators may be commercial nonreagent grade materials.

4.6.1 **Visual examination.** Each container shall be examined for conformance to section 5. Contractor certification verifying conformance to 3.2, 3.2.1, 3.2.2, 3.2.3 and 3.5 shall be required (see 6.2). Each tape roll shall be examined for conformance to 3.7.

4.6.2 **Uncured material tests.**

4.6.2.1 **Volatile content.**

a. Cut a 16 ± 2-square-inch specimen with a minimum width of 1.5 inches, and weigh to the nearest 0.001 gram (g) (W1).

b. Place specimen in recirculating oven preheated and stabilized to 325 ± 10°F for 10 ± 1 minutes. Specimens should be placed so that the specimens will get adequate air circulation.

c. Remove specimen, place in desiccator, and cool to room temperature.

d. Remove specimen from desiccator, and weigh to nearest 0.001 (W2).

e. Calculate percent volatiles for each specimen as follows:

$$\text{Percent volatiles} = \frac{W1 - W2}{W1} \times 100$$

Where: W1 = original weight of specimen, g

W2 = final weight of specimen after devolatilization, g

f. Record volatile content to nearest 0.1 percent.

4.6.2.2 **Resin solids, filler content, and cloth content.**

a. Place a 1.5 ± 0.1 g specimen weighed to the nearest 0.001 g (specimen weight) in 250 milliliter (ml) beaker.

b. Calculate devolatilized specimen weight (W) as follows:

$$W = \text{Specimen wt} - \frac{(\%\ \text{volatiles})\ (\text{specimen wt})}{100}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

Where: Percent volatiles is the value determined from the volatile content test (see 4.6.2.1)

c. Add 50 ml of N,N dimethyl formamide (DMF), and bring to boil. Gently stir boiling DMF and preimpregnated cloth for minimum of 5 minutes. Remove beaker from heat source, and cool for minimum of 5 minutes while stirring gently.

d. Decant and transfer liquid solution to a second 250 ml beaker.

e. Rinse remaining fabric with DMF several times to ensure complete extraction of resin and filler from fabric and add rinses to second 250 ml beaker.

f. Final rinse fabric with acetone to remove all traces of DMF.

g. Add all rinses to second 250-ml beaker.

h. Dry a filter and a filtering crucible at 350 ± 10°F for a minimum of 30 minutes. Weigh crucible and filter to nearest 0.001 g (W2).

i. Dry fabric in oven for 15 ± 1 minutes at 325 ± 10°F, cool in desiccator to room temperature, and reweigh to nearest 0.001g (W1).

j. Using an aspirator or vacuum pump, filter contents of second 250 ml beaker through filter and filtering crucible into vacuum flask.

k. Rinse crucible and filter several times with acetone to remove all resin.

l. Dry crucible and contents at 350 ± 10°F for 20 ± 1 minutes, cool in desiccator to room temperature, and reweigh to nearest 0.01 g (W3).

m. Calculate resin solids content, filler content, and cloth content as follows:

$$\text{Resin solids content, percent} = \frac{W - (W1 + W4)}{W} \times 100$$

$$\text{Filler content, percent} = \frac{W4}{W} \times 100$$

$$\text{Cloth content, percent} = \frac{W1}{W} \times 100$$

Where: W = calculated devolatilized weight of preimpregnated specimen, g

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

W1 = weight of fabric after resin and filler removal, g

W2 = crucible and filter weight, g

W3 = filler + crucible and filter weight, g

W4 = W3 - W2 = filler content, g

n. Record resin solids, filler content, and cloth content to nearest 0.1 percent.

4.6.2.3 **Resin flow.**

a. Using 4-inch by 4-inch ± 1/8 inch template, cut four squares across width of tape. All squares shall be bias cut to eliminate fiber loss in testing. Stack squares uniformly to make specimen.

NOTE: When retesting tape which is less than 4.0 inches for shelf life extension (see 3.6.1), it is permissible to use multiple tape widths which combine to provide a 4.0 inch specimen. Alternate tape layers shall be rotated 90 degrees.

b. Weigh specimen of 4 plies to nearest 0.01 g, and record as W1.

c. Place specimen between release film. Preheat press to 325 ± 10°F, position specimen in middle of press plate, and apply press load of 150 ± 10 psi immediately for minimum of 10 minutes at 325 ± 10°F.

d. Remove specimen from press, and cool to ambient temperature.

e. Using knife, scrape off resin flash to original size of specimen. Do not remove any reinforcement from original dimensions.

f. Reweigh specimen to nearest 0.01 g, and record as W2.

g. Calculate percent resin flow as follows:

$$\text{Percent flow} = \frac{(W1 - W2)}{W1} \times 100$$

Where: W1 = original weight of specimen, g

W2 = final weight of specimen, g

h. Record resin flow to nearest 0.1 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

4.6.3 Cured material tests.

4.6.3.1 Preparation of test panels. Prepare a test panel of sufficient size to provide specimens for all required tests as follows:

a. Cut sample material as required to fabricate panel as specified in individual test procedure, and cure plies into panel between separator film on smooth caul plates as follows:

(1) Orient plies such that the warp fibers of all plies are aligned in the same direction.

(2) Place stack of plies into press preset at 325 ± 10°F.

(3) Increase pressure slowly to 1000 ± 10 psi, allowing resin to stage (see 6.3.6), minimizing flash at ply edges.

(4) Hold pressure and temperature for 120 ± 15 minutes or one hour per inch thickness, whichever is greater.

(5) Decrease temperature and pressure to ambient.

NOTE: The outer 0.5 inch of the test panel shall not be used for preparation of specimens.

4.6.3.1.1 Density.

a. Cut sufficient plies to produce test panel 0.125 ± 0.025 by 4 inches (minimum) by 5 inches (minimum), mark warp direction, and prepare panel as specified in 4.6.3.1.

b. Cut specimen 1-inch by 1-inch by panel thickness at random from test panel. Wipe specimen clean using methyl ethyl ketone (MEK) conforming to ASTM D 740 and allow to dry in air 20 minutes minimum before testing.

c. Weigh and record weight of specimen in air to nearest 0.1 milligram (mg) (W1).

d. Weigh specimen in water at 73 ± 3°F, and record weight to nearest 0.1 mg using wire and sinker if required (W2).

e. Calculate density as follows:

$$\text{Density (g/cc)} = \frac{0.9975\ W1}{(W1 + W3 - W2)}$$

Where: W1 = weight of specimen in air, g

W2 = weight of specimen, wire, and sinker suspended in water, g

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

W3 = weight of wire and sinker suspended in water to the same depth as in W2 determination to nearest 0.1 mg, g

0.9975 = factor for converting specific gravity to g/cc

f. Record density to nearest 0.01 g/cc.

4.6.3.1.2 Residual volatile content.

a. Cut sufficient plies to produce test panel 0.250 ± 0.050 inch thick by 4.000 ± 0.125 by 4.000 ± 0.125 inches. Prepare panel as specified in 4.6.3.1 with the following exceptions:

Apply contact pressure for 30 ± 5 seconds and dump pressure for 2 cycles before slowly applying maximum pressure. (The material shall not be staged.)

b. Cut specimen 1.000 ± 0.050 inch by 1.000 ± 0.050 inch by panel thickness from center 2-inch by 2-inch section of test panel. Top and bottom molded surfaces shall not be machined. Wipe specimen clean using MEK conforming to ASTM D 740, and allow to dry in air 20 minutes minimum before testing.

c. Place specimen in desiccator, and desiccate for 18 hours minimum.

d. Weigh specimen to nearest 0.001 g, and record as W1.

e. Place specimen in air-circulating oven which has been preheated and stabilized at 325 ± 10°F for 30 minutes. Condition specimen at 325 ± 10°F for 4 to 4.5 hours.

f. Remove specimen from oven, and cool in desiccator for 30 ± 6 minutes or until specimen reaches room temperature.

g. Reweigh specimen to nearest 0.001 g. Record as W2.

h. Calculate percent residual volatiles as follows:

$$\text{Percent volatiles} = \frac{(W1 - W2)}{W1} \times 100$$

Where: W1 = original weight of specimen, g

W2 = devolatilized weight of specimen, g

i. Record residual volatiles to nearest 0.01 percent.

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:28Z
Check the source to verify that this is the current version before use.

4.6.3.1.3 **Resin content.** Resin content shall be determined in accordance with the following:

a. Prepare test specimens as follows:

1. Cut 32 rectangles (1/2 inch by 5 inches nominal) from a prepreg sample.

2. Stage the 32 rectangles at 255 ± 5°F to reduce flow.

NOTE: Staging time is dependent on the resin system and flow and must be determined from experience.

3. Using the 32 rectangles, mold a bar 1/2 inch by 1/2 inch by 5 inches.

4. Cure the specimen at 325 ± 10°F and 1000 ± 10 psi for 10 ± 1 minutes. If any resin flash, repeat and use longer staging time.

5. Cut the bar into approximately three equal lengths to form three test specimens.

b. Wipe specimen clean using MEK conforming to ASTM D 740, and allow to dry in air 20 minutes minimum before testing.

c. Record specimen weight to nearest 0.001 g (W).

d. Place test specimen in Vycor test tube and tightly attach tube to test apparatus (see figure 1).

e. Start vacuum pump, and evacuate system to 10 ± 5 millimeters (mm) of mercury (Hg) absolute. Check for air leaks by clamping off hose to vacuum pump and noting manometer. The leak rate must be less than 0.2 inch per minute. Recheck joints until this level is achieved.

f. With vacuum on and pressure stabilized at 10 ± 5 mm Hg, place the Fisher burner under the specimen. Adjust flame area to cover sample completely.

NOTE: A tube furnace maintained at 1500 ± 25°F may be used in place of the Fisher burners.

g. Continue pyrolysis for 30 ± 6 minutes or until manometer or gauge reading is same reading as prior to pyrolysis, whichever is longer.

NOTE: If residue collects in Vycor test tube, place Fisher burners under those areas until burnout is complete.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

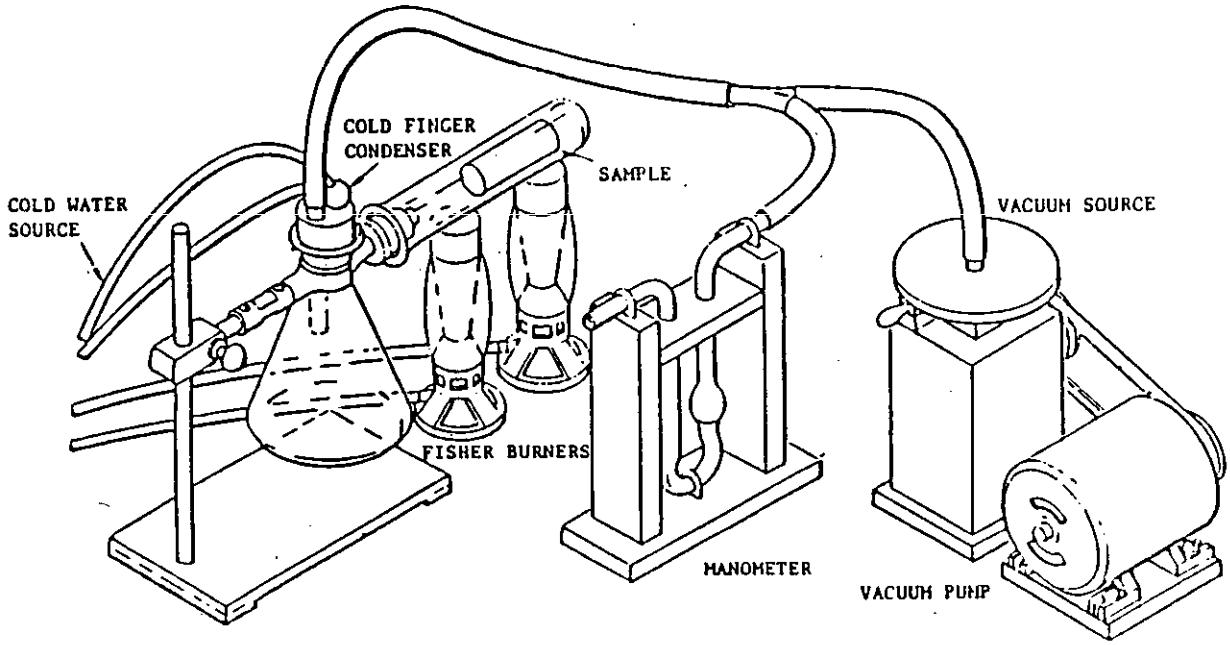


FIGURE 1. Resin content apparatus setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

h. When pyrolysis is complete, turn off the burners and cool under vacuum until the specimen is at room temperature (touch Vycor test tube just under test specimen).

i. When the specimen is cool, turn off vacuum and carefully let air into system. When pressure equilibrium is reached, remove the specimen from the test tube and weigh to nearest 0.001 g (W1).

j. Calculate resin content as follows:

$$\text{Resin content, percent} = \frac{(W - W1)}{W} \times 100\ (K)$$

Where: W1 = weight of specimen after pyrolysis, g

W = weight of specimen before pyrolysis, g

K = constant ratio of percent resin (Soxhlet extraction of DMF wash) to percent weight loss by pyrolysis (K for U.S. Polymeric = 2.008, K for Fiberite = 2.060)

k. Record resin content to nearest 0.1 percent.

4.6.3.1.4 **Compressive strength.** The compressive strength shall be determined, using the test panel prepared in 4.6.3.1.1, in accordance with the following:

a. Machine specimen to dimensions shown in figure 2.

b. Measure width and thickness of specimen to nearest 0.001 inch in gauge section area. Calculate and record cross-sectional area.

c. Place specimen in support jig (see Figure 4 of ASTM D 695), centered so ends project an even amount beyond both ends of jig. Tighten finger tight only.

d. Place specimen and support jig in testing machine, and align so specimen ends are parallel with surface of compression tool. Adjust crosshead of testing machine until it just contacts top of specimen.

e. The testing machine crosshead speed shall be 0.050 $\pm$ 0.010 inch per minute. Start machine.

f. Record maximum load carried by specimen during test.

g. Calculate compressive strength by dividing maximum load by original cross-sectional area.

h. Record compressive strength to nearest 10 psi.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

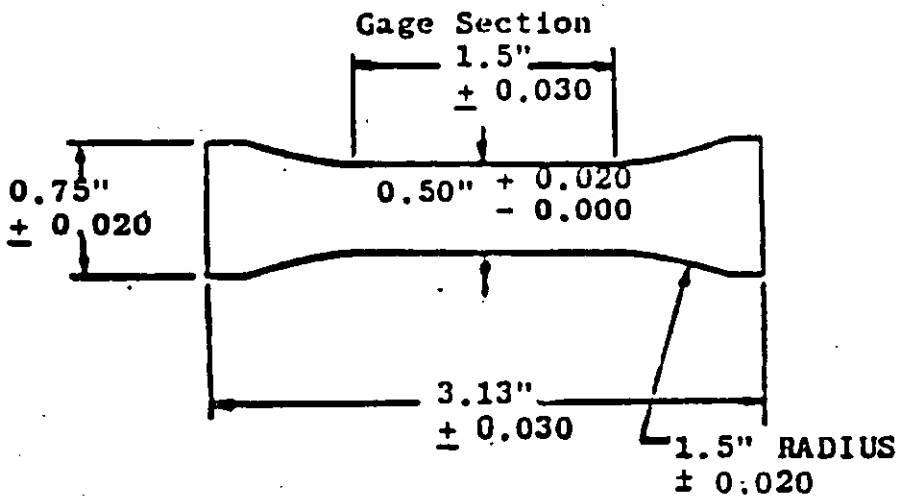


FIGURE 2. Compressive strength test specimen.

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:28Z
Check the source to verify that this is the current version before use.

4.6.3.1.5 Interlaminar shear.

a. Cut sufficient plies to produce test panel 1.50 ± 0.25 inch thick across plies by minimum dimensions of 1.5 inch by 1.5 inch.

b. Spray mold with release agent, and lay up plies into mold.

c. Cure as specified in 4.6.3.1.

d. After cure, remove mold from press, remove test panel from mold, and allow panel to cool to room temperature.

e. Cut a specimen of sufficient size to produce a 0.375-inch-diameter by 1.125-inch-length (minimum) finished specimen with length perpendicular to plies from the test panel.

f. Center the specimen to get length of cylindrical specimen perpendicular to plies, and machine the specimens to diameter of 0.375 +0.000, -0.002 inch.

g. Measure the diameter of test specimen to 0.0001 inch and calculate the cross-sectional area (A).

h. Place the specimen in a tensile loading shear jig conforming to figure 3.

i. Place shear fixture in a testing machine, align and pin the ends of fixture.

j. Set the machine at an indicated crosshead speed of 0.015 to 0.025 inch per minute, and test.

k. Record the maximum load carried by the specimen during test, and record as L.

l. Calculations: $\text{Interlaminar shear (psi)} = \frac{L}{2A}$

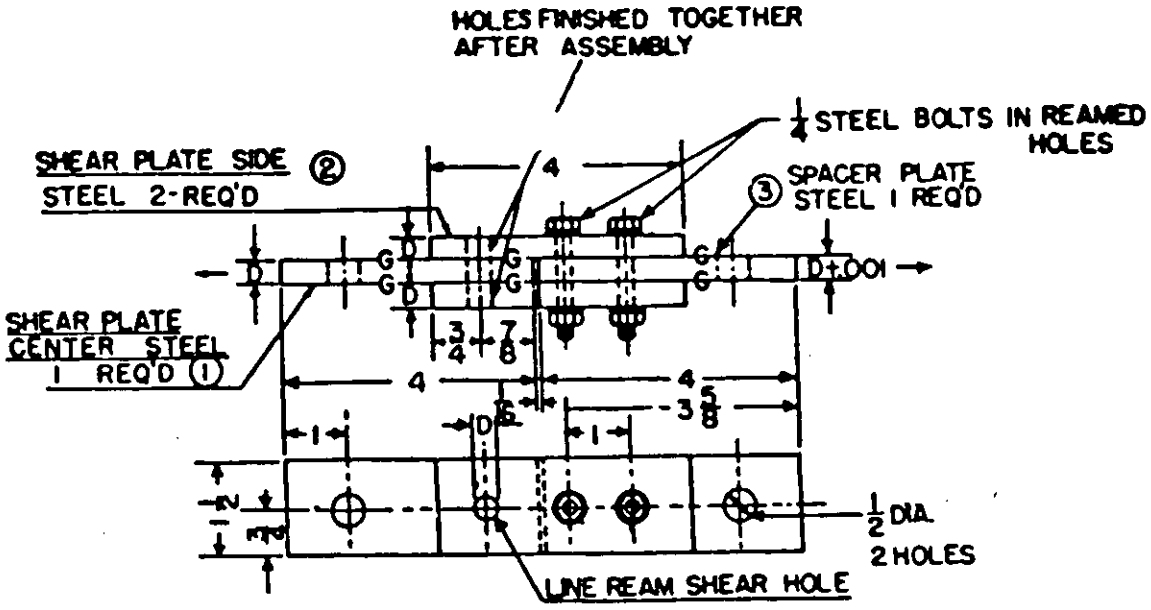
Where: L = maximum load carried by specimen during testing, in pounds

A = original cross-sectional area of specimen, in square inches

m. Record interlaminar shear to nearest 10 psi.

4.6.3.1.6 **Thermal conductivity.** Thermal conductivity shall be determined in accordance with ASTM C 177 and as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.



NOTES

1. D is nominal diameter of test cylinder.
2. Finish: machine after heat treatment.
3. Grind surfaces denoted G.
4. Spacer plate (3) to be 0.001 inch thicker than (1).
5. Edges of shear hole must be sharp.
6. All dimensions in inches; tolerances—fractions ± 1/64; specimen diameter shall be D + 0.000, — 0.001; diameter shear hole shall be D + 0.000, — 0.001.
7. Alinement of shear holes must be maintained.
8. Material: steel, heat-treated to Rockwell C 35–40.

FIGURE 3. Shear jig, tensile loading.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

a. Prepare a test panel of required size, and cure as specified in 4.6.3.1

b. Data shall be recorded for temperature of 250 ± 20°F.

c. The maximum test panel dimension shall be 9 inches. Wrinkles are not permitted in "with ply" specimen.

d. Record the thermal conductivity to nearest 0.01 Btu/(hr)(ft)(°F).

4.6.3.1.7 **Coefficient of thermal expansion.** The coefficient of thermal expansion (see 6.3.4) shall be determined in accordance with ASTM E 228 and as follows:

a. Prepare a test panel of required size, and cure as specified in 4.6.3.1.

b. Data shall be recorded for temperature range from 75 to 400°F.

c. The maximum test panel specimen length shall be 4 inches.

d. Record the coefficient of thermal expansion to nearest 0.010 in/in-°F X 10^{-6}.

4.6.3.1.8 **Flexural strength.** Flexural strength shall be determined in accordance with ASTM D 790, Method I, and as follows:

a. Prepare a test panel of the required size, and cure as specified in 4.6.3.1.

b. The maximum test specimen thickness shall be 0.25 inch.

c. Record the flexural strength to the nearest 10 psi.

4.7 **Inspection of packaging.** The containers and container markings shall be visually examined for compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein. Each tape roll shall be separately wrapped in a plastic bag.

5.1.1 **Level C.** The tape shall be packaged to afford adequate protection against loss, contamination, deterioration and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.6. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 Marking.

5.2.1 Standard marking. In addition to any special marking required by the contract or order (see 6.2), the interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 Special marking. In addition to the marking requirements of 5.2.1, each container marking shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Material trade name and identification.

d. Date of manufacture.

e. Number of rolls and roll net weight

f. Mark opposite sides of each shipping container with:

"TEMPERATURE SENSITIVE MATERIAL.
DO NOT EXPOSE TO TEMPERATURE HIGHER THAN 100°F."

g. Mark each roll of tape on inside of spool with contractor name, broadgoods roll number, tape roll number, lot number and material name with ink stamp, stencil or adhesive label.

h. Lot number and date of qualify conformance inspection (see 4.4).

i. Shelf life and inspection/test date for extension (see 3.6). In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 6 MONTH PERIOD PROVIDED THE TAPE, UPON RETEST, CONFORMS TO THE RESIN FLOW AND VOLATILES CONTENT REQUIREMENTS SPECIFIED IN MIL-T-82839."

5.3 Shipment temperature. The temperature of the tape during shipment shall not exceed 100°F as measured by temperature-indicating tape placed on the inside of the spool of at least one tape roll in each container in each shipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The tape specified herein is intended for use in the manufacture of Standard Missile MK 104 Dual Thrust Rocket Motor nozzles.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, revision letter, and date of this specification.

b. Whether first article is required (see 3.1 and 6.4).

c. Packaging required, if other than as specified in 5.1.

d. Special marking if other than as specified in 5.2.2.

e. DOD FAR 52.246-15 (see 4.6.1).

f. Tape width.

g. Acceptance stock and lot number.

6.3 Definitions

6.3.1 Broadgoods roll. A wide cloth roll received from the cloth contractor and impregnated with a resin.

6.3.2 Specimen. A portion of the material (as is, or further processed) selected from a sample roll for inspection.

6.3.3 Tape roll. A narrow tape cut from a product roll of impregnated cloth.

6.3.4 Coefficient of thermal expansion. Coefficient of thermal expansion is the secant slope of the linear expansion between 75 and 400°F.

6.3.5 Sample. A broadgoods roll of impregnated cloth selected in accordance with the requirements of this specification for performance of inspection.

6.3.6 Stage. A method of stabilizing and retaining the phenolic polymer in a solid state on the carbon cloth by temperature cycling.

6.3.7 Bias. The angular orientation of the warp and fill of the cloth weave relative to the longitudinal axis of the cut tape.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

6.4 **First article.** When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item (s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instruction in acquisition documents regarding arrangements for examinations, approval of first article test regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 **Subject term (key word) listing.**

Nozzles, rocket
Rocket Motor, Dual Thrust, MK 104
Standard Missile

6.6 **Supersedure information.** MIL-T-82839 incorporates the following engineering change proposal (ECP) and notice of revision (NOR):

ECP		NOR	
MTA062	3/30/88	MTA062.2	3/30/88

Preparing Activity:
Navy-OS
(Project 1340-N811)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-82839	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
TAPE, CARBON CLOTH, PHENOLIC-PREIMPREGNATED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-03-23T07:28Z
Check the source to verify that this is the current version before use.